



ENVT. MGT. Department
SIMHADRI

Note No :1



Ref:SMTPP/ENVT. MGT./2025-26/EMG-AU/387289

Date:

18/08/2025(17:18:33)

Subject:LMI on Control of Liquid Effluents from Simhadri Super Thermal Power Station

Competent authority had accorded approval vide notesheet :**SMTPP/ENVT. MGT./2024-25/EMGAU/713428** dated **31.01.2025** (Copy enclosed) for the approval of LMI on Control of Liquid Effluents from Simhadri Super Thermal Power Station Rev-09.

However, a revised OGN is issued with following revisions on 28th July 2025. (Copy Enclosed)

Records of revision in OGN

- Addition of NTPC water policy & Rainwater Policy in **Section 4.12**.
- Addition of **Section 5.5** which mandates that Monitoring of water consumption and effluent generation for plant/Township area shall be done as per stipulated quantity in Consent to operate.
- Addition of **Section 6.6** which mandates that All effluents generated from area of boiler, turbine, CHP and other of main plant shall be collected in Oil Settling pits located at designated areas.
- Addition in **Section 6.7** of the provision that effluent from other sources, such as FGD, shall be routed as per the engineering scheme provided for station.
- Modification in **Section 8.3** which includes the provision of Offline pH monitoring facility in addition to the Online pH monitoring facility at CSSP of stations where coal beds are prepared with mill rejects. Record of Deviation % to be recorded in the last sampling for validation/ verification.
- Modification in **Appendix-C (D)** which mandates Condition of the zero liquid discharge shall be complied without any deviation, as per company target. Also, the discharge point shall preferably be as per stipulated conditions of EC/CTO/SPCB/CPCB direction/guidelines.

Accordingly, the LMI has been revised and put up for review and approval please.

Submitted for approval please.

Submitted by:

सिबिन अब्राहम - वरिष्ठ प्रबंधक

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On:18/08/2025(17:18:33) **Ref:**SMTTP/ENVT. MGT./2025-26/EMG-AU/387289

Note No:2

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Note No:3

Please review.

Submitted by:

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Note No:4

Reviewed

Submitted by:

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On:28/08/2025(17:53:18) **Ref:**SMTTPP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:5

Following Para of OGN needs to be incorporated in the LMI:

- Para 4, & 9-12 of 5.0
- Para-A (1-4 & 7) and Para-B of 6.0
- Some Para of 7.0 are missing in the LMI.
- Appendix-A: Minimum, Maximum and Average Values are required.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

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On:30/08/2025(10:50:34) **Ref:**SMTTPP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:6

Submitted by:

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Note No:7

All the suggestions and recommendations highlighted above as per the latest OGN have been incorporated in the updated LMI in the attachment. Para A and Para B of 6.0 have been merged and have been modified w.r.t to the Simhadri Station.

Submitted by:

सिबिन अब्राहम - वरिष्ठ प्रबंधक

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On:16/09/2025(13:19:52) **Ref:**SMTPP/ENVT. MGT./2025-26/EMG-AU/387289

Note No:8

LMI is in line with the OGN. Forwarded for further review.

Submitted by:

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On:17/09/2025(09:53:22) **Ref:**SMTPP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:9

May please check and comment.

Submitted by:

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On:18/09/2025(15:38:08) **Ref:**SMTPP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:10

The LMI is in line with OGN.

Points related to Chemistry are being complied.

Submitted by:

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Note No:11

Submitted for approval please.

Submitted by:

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On:19/09/2025(10:56:51) **Ref:**SMTTP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:12

Forwarded for further review and approval.

Submitted by:

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Note No:13

May pls review.

Submitted by:

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On:24/09/2025(09:59:59) **Ref:**SMTTP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:14

Reviewed and submitted for further process.

Submitted by:

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Note No:15

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On:26/09/2025(11:47:31) **Ref:**SMTTPP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:16

LMI reviewed and responsibilities noted.

Forwarded for further approval please.

Submitted by:

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Note No:17

Reviewed. File may be processed further.

Submitted by:

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Note No:18

Reviewed. File may be processed further.

Submitted by:

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Note No:19

Forwarded for review

Submitted by:

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On:27/09/2025(10:29:44) **Ref:**SMTTPP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:20

LMI reviewed and forwarded for further approval.

Submitted by:

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On:29/09/2025(11:58:02) **Ref:**SMTPP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:21

Forwarded for further review and approval.

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

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Note No:22

Forwarded for further review and approval.

Submitted by:

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Note No:23

Reviewed and forwarded.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

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Note No:24

Reviewed and is being forwarded for further process please.

Submitted by:

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Note No:25

LMI is in line with the revised OGN. May please be approved.

Submitted by:

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On:03/10/2025(19:44:19) **Ref:**SMTTP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:26

Submitted by:

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Note No:27

Submitted by:

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On:04/10/2025(15:27:50) **Ref:**SMTTP/ENV.T. MGT./2025-26/EMG-AU/387289

Note No:28

Approved & Submitted by:

समीर शर्मा - कार्यकारी निदेशक

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On:04/10/2025(16:48:20) **Ref:**SMTTP/ENV.T. MGT./2025-26/EMG-AU/387289

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**Control of Liquid Effluents from
Simhadri Super Thermal Power Station**

LMI/OGN/ENV/008 Rev: 10

AUG-2025

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	THE CONTROL OF LIQUID EFFLUENTS FROM SIMHADRI SUPER THERMAL POWER STATION
2	LMI No.	LMI/OGN/ENV/008 Rev:10
3	Reference Document	COS-ISO-00-OGN/ENV/008 / Rev:04 28 th July 2025
4	SupersededDocument	LMI/OGN/ENV/008 Rev:09 DEC 24
5	Prepared By	Name: Sibin Abraham, Senior Manager (EMG & AU)
6	Checked By	Name: Rohit Singh DGM (EMG & AU)
7	Reviewed By	HOD (Chem)
		HOD (CHP)
		HOD (Civil)
		HOD (FGD)
		HOD (EEMG)
		HOD (C&I)
		HOD (AHP)
		HOD (Maint.)
		HOD (Operation)
		HOD (Medical)
		HOD (O&M)
8	Approved By	H O P (Simhadri)

TITLE	The Control of Liquid Effluents from Simhadri Super Thermal Power Station
LMI No.	LMI/OGN/ENV/008 Rev:10

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TITLE	The Control of Liquid Effluents from Simhadri Super Thermal Power Station
LMI No.	LMI/OGN/ENV/008 Rev:10

1.0 Introduction

This LMI describes important requirements to control the liquid effluents discharged from Simhadri Super Thermal Power Station and gives guidance on complying with legislation and environmental norms on control of liquid effluents and their quality.

Such measurements are necessary to provide an absolute value with manual measurement and as mandatory Effluent Quality Monitoring System (EQMS) to demonstrate compliance against a specific standard or the requirements of consent orders by APPCB/CPCB/MoEF & CC or as stipulated in NOC/EC/CTO/ orders of the court/NGT. Inter comparison of manual and EQMS shall be done on quarterly/regular basis in addition to calibration of the both types of monitoring system and for the occasional requirement.

2.0 Superseded Documents

LMI/OGN/ENV/008 REV.09 Date: Dec 2024

3.0 Details of Revision

Following are the proposed revisions as per the change in the OGN

- Addition of NTPC water policy & Rainwater Policy in Section 4.0(12).
- Addition of Section 5.5 which mandates that Monitoring of water consumption and effluent generation for plant/Township area shall be done as per stipulated quantity in Consent to operate.
- Modification in Section 8.3 which includes the provision of Offline pH monitoring facility in addition to the Online pH monitoring facility at CSSP of stations where coal beds are prepared with mill rejects. Record of Deviation % to be recorded in the last sampling for validation/ verification.
- Modification in Appendix-C (D) which mandates Condition of the zero liquid discharge shall be complied without any deviation, as per company target. Also, the discharge point shall preferably be as per stipulated

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conditions of EC/CTO/SPCB/CPCB direction/guidelines.

- Addition in Section 6.9 of the provision that effluent from other sources, such as FGD, shall be routed as per the engineering scheme provided for station.
- Addition of Section 6.6 which mandates that All effluents generated from area of boiler, turbine, CHP and other of main plant shall be collected in Oil Settling pits located at designated areas.

4.0 Introduction to the legal framework

The law relating to the management and control of liquid effluents is basically governed by the following Acts and Rules:

- 4.1** The Environment (Protection) Act 1986 and amendments thereof
- 4.2** Water (Prevention & control of pollution) Act,1974 and amendments thereof
- 4.3** Water (Prevention & control of pollution) rule,1975
- 4.4** Environment (Protection) Amendment,2015
- 4.5** The Hazardous and other wastes (Management and Transboundary movement) Rules,2016
- 4.6** Bio-Medical Waste Management Rules, 2016 and amendment thereof.
- 4.7** Directions under Section 18(1)(b) of the Water (P&CP) Act, 1974 regarding continuous Monitoring of industrial effluent quality before final discharge into river Ganga and its tributaries (Ganga River Basin) for self-monitoring of compliance.
- 4.8** Directions under Section18(1)(b) of The Water(P&CP) Act 1974 and The Air(P&CP) Act 1981 in the matter of Pollution control in 17 Categories of highly polluting Industries, CETPs and common Hazardous waste & Bio- Medical Waste Incinerators regarding self- monitoring of Compliance.
- 4.9** MoEF& CC notification dated 13.10.2017 for Effluent discharge standards by STP (sewage treatment plants).
- 4.10** NGT order dated 30.04.2019 (Nitin Shankar Deshpande vs Union of India & others) OA no 1069/2018 for STP standards.
- 4.11** Waste Management Policy 2022 & Zero Deviation Document.

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4.12 NTPC Water Policy & Rainwater Harvesting Policy.

5.0 General guidelines for management of effluents

The following are a few general guidelines for management of liquid effluents discharged from thermal power stations. Simhadri Super Thermal Power Station ensured drain separation (effluent & Storm water) and Zero Liquid Discharge from Plant area, Ash Dyke area and Township area.

5.1 The station should have valid Consent Orders for Trade Effluents /Domestic Effluents.

Res: HOD(EMG)

5.2 An application for renewal of the Consent Order is to be submitted to the Andhra Pradesh State Pollution Control Board (APPCB) at least 60 days before the expiry of existing Consent Order.

Res: HOD(EMG)

5.3 State Pollution Control Board be informed about any change in effluent discharge with respect to location, quality, quantity or addition of new discharge point.

Res: HOD(EMG)

5.4 Trade effluent drain and storm/ runoff water drain shall not be allowed to mix either within boundary or outside boundary in all seasons including rainy season.

Res: HOD (OPN)

5.5 Monitoring of water consumption and effluent generation for plant/Township area shall be done as per stipulated quantity in Consent to operate.

Res: HOD(EEMG)

5.6 All effluents containing pollutants above the statutory limits are to be treated in a Central Monitoring Basin in the Effluent Treatment Plant (ETP).

Res: HOD (CHEM)

5.7 Monitoring is to be done regularly for the effluent leaving the plant boundary at common monitoring basin.

Res: HOD (CHEM)

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5.8 All attempts must be made for conservation of water - a scarce natural resource. Wherever and as far as possible, effluent re-circulation systems shall be adopted.

Res: HOD (OPN) & HOD(EEMG)

5.9 Sample from plant main fall out drain (effluent (s) water which is (are) going outside plant boundary) to be regularly analyzed and monitored for pH and other parameters and ensure compliance.

Res: HOD (CHEM)

5.10 All effluent must be subject to Primary/secondary/ tertiary treatment as required to meet the norms applicable for respective type of effluent. Treated effluent shall be reused for activities designated for reuse at station to reduce the quantity of water drawl from the main water source.

Res: HOD(CHEM) & HOD(OPN)

5.11 Frequent periodic (quarterly) tests for contamination of surface water and ground water must be done for toxic contaminants and heavy metals such as mercury, arsenic, cadmium etc. mentioned in IS 10500-2012 beside the other legal requirements.

Res: HOD(CHEM)

5.12 Wastewater needs to be treated to reduce suspended solids, biodegradable organics, etc. to acceptable levels prior to disposal.

Res: HOD (CHEM)

6.0 Possible origins and nature of liquid effluents in Simhadri thermal power station

The type and nature of liquid effluents discharged from Simhadri Super Thermal Power Station are given below which must be treated or attended by concerned departments as indicated below and corresponding record of actions should be maintained.

TITLE	The Control of Liquid Effluents from Simhadri Super Thermal Power Station
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- 6.1** Effluents generated from Water Treatment Plant (WTP) - either acidic / alkaline neutral in nature, shall be pumped to location of designated reuse (ash slurry/ bottom ash slurry system as the case may be) after treatment / neutralization.

Res: HOD(CHEM)

- 6.2** Domestic and sanitary effluents from plant and township areas shall be connected to STP with arrangement of disinfectant after treatment. Treated STP effluent meeting all parameters as per the norms applicable for treated STP effluent & after proper disinfectant shall be used in horticulture/ afforestation/agriculture.

Res: HOD(CIVIL)

- 6.3** Effluents Generated from Boiler Blowdown which may contain impurities from Boiler Drum shall be used in ash handling/ service water.

Res: HOD (OPN)

- 6.4** Condenser cooling water (CCW) discharge or Cooling Tower blow-down shall be used in ash handling/ service water as per station practice.

Res: HOD (OPN)

- 6.5** Occasional miscellaneous effluents from plant such as floor washings, storm water drains / run-off water etc. (may contain materials of generally suspended / floating / oily / greasy type) may be sent to the subsystem designated for reuse.

Res: HOD(OPN) & HOD(MAINT.)

- 6.6** All effluents generated from area of Boiler, Turbine, CHP and other of main plant shall be collected in Oil Settling pits located at designated areas.

Res: HOD(MAINT.)

- 6.7** Effluent generated from hospital shall be treated with hypo in a separate tank as per the BMW Rule 2016 and after treatment.

Res: HOD (MEDICAL)

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6.8 Decanted water from ash lagoons collected in OFL and Toe drain water (all dykes & OFL) may contain suspended ash particles and shall be recirculated by AWRS and reused in ash slurry system and dust suppression in ash dyke only.

Res: HOD(OPN) & HOD(AHP)

6.9 Run-off and decanted Effluents from coal handling area (may contain coal articles) shall be neutralized and connected through dedicated drains to CSSP for further treatment.

Res: HOD(CHP)

6.10 Effluent from other sources, such as FGD shall be routed as per engineering scheme.

Res: HOD (FGD)

7.0 Treatment practices for water effluents

Treatment practices being followed for water effluents emanating from Simhadri Thermal Power Station are given below.

7.1 All effluent must be subjected to Primary/secondary/ tertiary treatment as required to meet the norms applicable for respective types of effluent. Treated effluent shall be reused for activities designated for reuse at the station to reduce the quantity of water drawn from the main water source.

Res: HOD (OPN) & HOD(CHEM)

7.2 Periodic tests for contamination of surface water once in 06 months and ground water once in 03 months must be done for toxic contaminants and heavy metals such as mercury, arsenic etc.

Res: HOD(CHEM)/HOD (EMG)

7.3 Wastewater needs to be treated to reduce suspended solids, biodegradable organics, etc. to acceptable levels prior to disposal.

Res: HOD(CHEM)

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7.4 Decanted Effluents from coal handling area (DE, DS and sprinkler system in Coal stacking Yard), Mill area/Transfer points area and Mill reject storage area - may contain coal particles or low pH water, shall be neutralized and connected through dedicated drains to CSSP for further treatment, recirculation and reuse.

Res: HOD(CHP)

7.5 Provision may also be made to avoid carry-over of coal particles in the effluent from the top of the surface. Use of settled coal particles along with recycling and reuse of coal settling pit effluent may be ensured. Record of cleaning and service of coal settling pit need to be maintained.

Res: HOD (CHP)

7.6 Effluent from coal stack yard drains, CSSP as well as mill reject storage area will be regularly analyzed for pH in addition to other parameters. Any effluent with low pH (<6.5) shall be reused/diverted to other effluent stream/ sub systems (ash slurry system/ CSSP) only after neutralization. Storage of mill reject in multiple locations to be avoided.

Res: HOD(CHEM) & HOD(CHP)

7.7 Effluents from fuel oil handling area shall be passed through oil traps to avoid escaping of spilled over oil into final drains. This separation is achieved by density difference. Provision of oil skimmers may also be made for more effective separation of low concentration oil from the effluent.

Res: HOD (MAINT.)

7.8 All acidic or alkaline effluents from the plant such as effluents from Water treatment Plant, including effluents discharged during boiler cleaning operation etc. are routed through Neutralization Pits for their neutralization before discharge into ash water sump.

Res: HOD (OPN) & HOD (CHEM)

7.9 All domestic and sanitary effluents where biodegradable substances are stabilized aerobically from plant / township area, sludge generated is collected through clariflocculators in the Sewage Treatment Plant and discharged only after proper treatment and analysis. In no case, untreated effluent is to be discharged into natural water body. The treated water is being used for horticulture to the extent possible.

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Res: HOD (CIVIL)

7.10 Sufficient retention time shall be given to ash pond effluent in the ash pond for settlement of ash particles. Ash slurry discharge points should be changed periodically to avoid channeling.

Res: HOD (AHP)

7.11 Metering of recycled quantity of water shall be ensured to demonstrate the recirculated quantity of water.

Res: HOD(EEMG)

7.12 No water shall be discharged from OFL to any nearby water stream or land except in inevitable circumstances. If there is any incidental discharge from OFL to water body, detail shall be informed to concerned regulator.

Res: HOD(OPN) & HOD(EMG)

7.13 It should be encouraged to work on higher COC so that quantity of water requirement for make-up can be reduced.

Res: HOD (EEMG) & HOD(OPN)

8.0 Monitoring guidelines and methodology for effluents

The following are a few guidelines for monitoring of liquid effluents emanating from Power Station.

8.1. Flow is to be measured and recorded (through integrators) at all water intake points and Main plant effluent discharge points (this is a statutory requirement as per Water cess Act and is one of the consent conditions of Water Consent Order).

Wherever such provisions do not exist and till such time these provisions are made, portable instruments such as water current meter etc, may be used or, alternatively, flow may be computed on the basis of pump characteristics, pump running hours etc.

Res: HOD (EEMG.)

8.2 Monitoring of water at intake and effluent leaving plant boundary is done as per the formats (Appendix 'A'). The frequencies and parameters given in the formats are the minimum monitoring schedule followed. Any increase in frequencies or parameters as stipulated by State or Central regulating agencies from time to time

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is to be done additionally. The monitoring is to be done through an MOEF recognized laboratory.

Res: HOD (CHEM)/HOD (EMG)

- 8.3** The provision of on-line/Offline pH monitoring facility at CSSP of stations where coal beds are prepared with mill rejects and in ash Dyke also if mill rejects are used for construction of rock toe/toe drain etc.

Res: HOD (CHEM)

- 8.4** All efforts are to be made to comply with the conditions specified in the Water Consent including quality and quantity limits. All operations of the station must conform to the existing consent conditions or revised consent conditions as stipulated from time to time by the Andhra Pradesh State Pollution Control Board. Wherever gaps exist and it is not possible to comply with a qualitative or quantitative condition either because of system non-availability or inadequacy / deterioration of existing system, then suitable remedial action is to be taken to the extent possible.

Res: HOD (O&M)/HOD (EEMG)/HOD (EMG)

- 8.5** Every month, station In-charge of the Environment Management Group (EMG) shall generate Report with respect to quality, quantity and monitoring program (frequency / parameters) and put up to the corporate environment through the PARIVESH developed by corporate EMG.

Res: HOD(EMG)

- 8.6** The online EQMS parameters are to be monitored for exceptions.

Res: HOD(EMG)

- 8.7** Monthly report shall be generated as per the format (Appendix-E) and shall be communicated to CC-EMG and RHQ by In-charge of the Environment Management Group (EMG). All the effluent parameters given in the formats circulated by Corporate Environment Management shall be monitored using standard sampling and test methodologies as specified in Consent Order. Wherever not specified, methodology for sampling/testing as per IS-2488(Parts 1,2,3,4,5), IS-4733, IS3550, IS-3025 etc. may be used. Any other standard test method prescribed by WHO, ASTM, APHA, BS etc. may be used for parameters if so specified by a regulatory body or where there is no corresponding IS method for the same.

Res: HOD(EMG)

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9 Effluent Quality Monitoring System

Two Online effluent quality monitoring system i.e. EQMS-1 and EQMS-2 are available for the parameters prescribed by APPCB. Monthly Calibration of the EQMS analyzers are to be carried out. PTZ camera is installed at the Central Monitoring basin (CMB). All effluent quality parameters mentioned in CTO and equipment availability details are uploaded in Parivesh in the format Annexure-E.

Res: HOD(C&I)/ HOD(EMG)

10 Inter Comparison of EQMS and Manual monitoring data

Inter-comparison of EQMS and manual monitoring shall be done on quarterly basis. Whenever work undertaken for inter-comparison of manual monitoring data and EQMS data, If there is difference of more than $\pm 10\%$ in the data of manual monitoring and EQMS data, calibration of the both manual monitoring equipment and EQMS shall be taken up immediately and variation in both results shall be sorted out or the reason of the variation shall be mentioned in the monthly report with technical reasons.

Res: HOD(EMG)

11 Associated documents

- 11.1** Acts and Rules on Water (P&CP) Act, Air (P&CP) Act, EPA
- 11.2** Conditions stipulated in the NOC/EC/CTE &CTO by concerned regulatory or statutory bodies.
- 11.3** The Control of Environmental Pollution - Operation Guidance Note: COS-ISO-00OGN/ENV/001 Rev.03)
- 11.4** IS documents as listed in Appendix-C.
- 11.5** All related documents as per the legal framework (SI No 4.0)

12 Review

The HOD (EMG) will be responsible for reviewing this document in every two years or as necessary.

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APPENDIX –‘A’**ENVIRONMENT PERFORMANCE REPORT**

STATION NAME:

FORM-2 (EFFLUENTS)

PERIOD: MONTHLY

S. No.	Name of the Effluent Particulars	pH	TSS (mg/l)	Oil & Grease (mg/l)	Other parameters (P) as per CTO (Name of the parameter along with unit of measurement is to be given in place of P1, P2, etc)						Reasons for Deviation and Action Taken
					P1	P2	P3	P4	P5	P6	
1.	<u>Ash Pond Effluent</u>										
a	Minimum										
b	Maximum										
c	Average										
d	Total no. of readings										
e	No. of Readings Outside Limit										
2.	<u>Cooling Water Discharge/C.T. Blowdown</u>										
a	Minimum										
b	Maximum										
c	Average										
d	Total no. of readings										
e	No. of Readings Outside Limit										
3.	<u>Main Plant Effluent</u>										
a	Minimum										
b	Maximum										
c	Average										
d	Total no. of readings										
e	No. of Readings Outside Limit										
4.	<u>Domestic Effluent (Colony)</u>										
a	Minimum										
b	Maximum										
c	Average										
d	Total no. of readings										
e	No. of Readings Outside Limit										

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*TSS OR DIFFERENTIAL TSS, AS APPLICABLE (in ppm) AS PER CONSENT ORDER

TEMPERATURE DIFFERENTIAL BETWEEN AMBIENT WATER AND EFFLUENT.

\$STAGE WISE DATA IF DISCHARGED SEPARATELY.

NR PARAMETER NOT REQUIRED FOR THE EFFLUENT INDICATED.

DELETE WHICH EVER IS NOT APPLICABLE.

DELETE WHICH EVER IS NOT APPLICABLE; IF THERE IS MORE THAN ONE DOMESTIC EFFLUENT, PLEASE PROVIDE DATA FOR EACH EFFLUENT.

APPENDIX - 'B'

EP RULES, 1986 (SCHEDULE - I) Effluent Standards for Thermal Power Plants

S.No	Type of Effluent	Parameter	Maximum limiting concentration, milligram per liter (except for pH and temperature)
1.	Boiler Blow Down	Suspended solids Oil and Grease Copper (Total) Iron (Total)	100 20 1.0 1.0
2.	Cooling Tower Blowdown	Free Available Chlorine Zinc Chromium (Total) Phosphate Other Corrosion Inhibiting Materials	0.5-1.0 0.2 5.0 Limit to be established on case to case basis by Central Board in case of Union Territories and State Board in case of States
3.	Ash Pond Effluent	pH Suspended Solids Oil and Grease	6.5-8.5 100 20

*Revised Temperature Limits as per Environment (Protection) (second Amendment) Rules, 1998 as shown at Appendix - 'D'.

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APPENDIX – ‘C’

**ENVIRONMENT (PROTECTION) (SECOND AMENDMENT) RULES,
1998, G.S.R. –7, DATED 22.12.1998 PUBLISHED IN THE GAZETTE
OF INDIA NO. 1 PART– I, SECTION 3, SUB- SECTION (I), DATED
02.01.1999.**

(EXTRACTS OF THE RELEVANT SECTIONS OF THE NOTIFICATION
PERTAINING TO ENVIRONMENTAL STANDARDS FOR LIQUID
EFFLUENTS)

**LIQUID – WASTE DISCHARGE LIMITS FOR GAS / NAPHTHA – BASED
THERMAL POWER PLANTS (TPPs).**

Parameter	Maximum Limit of Concentration (mg /l except for pH and Temperature)
pH	6.5 – 8.5
Temperature	As applicable for other Thermal Power Plants
Free available chlorine	0.5
Suspended solids	100.0
Oil and grease	20.0
Copper (total)	1.0
Iron (Total)	1.0
Zinc	1.0
Chromium (total)	0.2
Phosphate	5.0

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**ENVIRONMENT (PROTECTION) (SECOND AMENDMENT) RULES, 1998,
G.S.R. -7, DATED 22.12.1998**

**1. TEMPERATURE LIMIT FOR DISCHARGE OF CONDENSER
COOLING WATER FROM THERMAL POWER PLANTS(TPPs).**

A) NEW TPPs COMMISSIONED AFTER 1.6.1999.

New thermal power plants, which will be using water from rivers/lakes/reservoirs, shall install cooling towers-irrespective of location and capacity. Thermal power plants which will use sea water for cooling purposes, the condition below will apply

B) NEW TPPs IN COASTAL AREAS USING SEAWATER

The thermal power plants using sea water should adopt suitable system to reduce water temperature at the final discharge point so that the resultant rise in the temperature of receiving water does not exceed 7 °C over and above the ambient temperature of the receiving water bodies.

C) EXISTING THERMAL POWERPLANTS

Rise in temperature of condenser cooling water from inlet to the outlet of condenser shall not be more than 10°C.

D) GUIDELINES FOR DISCHARGE POINT

1. Condition of the zero liquid discharge shall be complied without any deviation, as per company target and as per conditions stipulated in EC/NOC/CTE /CTO or by any directive/ court order etc.
2. In case of any deviation from the condition of ZLD, incident shall be reported to concerned regulator, ED(OS) and HOD (Env. Management-CC) without failure within 48 hours with detail physico-chemical analysis of water quality for parameter prescribed for TPP.
3. The discharge point shall preferably be located at the bottom of the water body at mid-stream for proper dispersion of thermal discharge or as per stipulated conditions of EC/CTO/SPCB/CPCB direction/guidelines.

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4. In case of discharge of cooling water into sea, proper marine out-fall shall be designed to achieve the prescribed standards. The point of discharge may be selected in consultation with concerned regulator /NIO.
5. No cooling water discharge shall be permitted in estuaries or near ecologically sensitive areas such as mangroves, coral reefs /spawning and breeding grounds of aquatic flora and fauna.
6. Any discharge from plant area shall be from the authorized point of discharge on which EQMS are installed and online data is being monitored and communicated to concerned regulators.
7. Direction issued by CPCB/SPCB for installation of EQMS and TPZ camera and data communication to the concerned regulator shall be ensured.

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APPENDIX - 'D'**LIST OF INDIAN STANDARDS FOR EFFLUENTS****1.0 TERMINOLOGY.**

- 1) IS-7022(Pt-1):1973- Glossary of terms relating to water, Sewerage and Industrial Effluents (Part-1)
- 2) IS - 7022 (Pt-2): 1979 - Glossary of terms relating to water, Sewerage and Industrial Effluents (Part-2)
- 3) IS - 10446: 1983 Glossary of terms for water supply and Sanitation.
- 4) MoEF& CC notification dated 13.10.2017 for Effluent discharge standards by STP(sewage treatment plants)
- 5) NGT order dated 30.04.2019 (Nitin Shankar Deshpande's Union of India & others) OA no 1069/2018 for STP standards

2.0 STP discharge standards as per NGT order 30.04.2019

S. No.	Type of Effluent	Parameter	Maximum limiting concentration, (milligram per litre)
	STP treated water	pH	5.5-9.0
		BOD	Less than 10
		Total Suspended solids (TSS)	Less than 20
		COD	Less than 50
		Total Nitrogen	Less than 10
		Dissolved P (Phosphorus)	Less than 1
		Faecal Coliforms, MPN/100ml	Less than 230

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Appendix – E**Online Effluent Quality Monitoring System****Name of the Station:**

Manual Monitoring Frequency of parameter :

Last Calibration of EQMS :

Comparison with Manual Monitored data :

Online percentage availability EQMS data in current FY :

Online data connectivity status to SPCB/CPCB :

Auto diagnostic system access to SPCB/CPCB :

Availability of the Camera installed at discharge, if applicable :

Exception in the EQMS availability, if any :

Name of the Person responsible for EQMS :

Contact Number & E-mail ID :

Name of the person & his contact detail communicated to
CPCB/SPCB/CC on (Date) :